

Guanghui Min

CONTACT INFORMATION

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RESEARCH INTERESTS

My research focuses on efficient information use in large language models, from data-centric post-training to context management for long-horizon agents. I work on data selection, synthesis, and curation for LLM post-training, as well as methods for preserving and leveraging task-relevant context over extended agent interactions. My broader interests include efficient machine learning, graph learning, and time-series modeling.

EDUCATION

2024/08 – Present	University of Virginia <i>Ph.D. in Computer Science</i> Advisor: Prof. Chen Chen	Charlottesville, VA
2018/09 – 2020/05	University of Michigan, Ann Arbor <i>M.Sc. in Applied Statistics</i>	Ann Arbor, MI
2014/09 – 2018/06	Wuhan University <i>B.Sc. in Statistics</i> Major GPA: 3.76/4.00; Rank: 6/47	Wuhan, China

PUBLICATIONS

Preprints & Under Review

- [1] **Guanghui Min**, Liang Wu, Mingjia Shi, Yinhan He, Mayank Darbari, Liangjie Hong, and Chen Chen. Adapting Context Compression for Long-Horizon Agents with Counterfactual Continuations. *arXiv preprint*, 2026.
- [2] **Guanghui Min** and Chen Chen. Prune the Summary, Not the History: Source-Judged Compaction for Long-Horizon Agents. *Under Review*, 2026.
- [3] Zaiyi Zheng, Liang Wu, **Guanghui Min**, Yaochen Zhu, Liangjie Hong, Chen Chen, Jundong Li. SAPO: Step-Aligned Policy Optimization for Reasoning-Based Generative Recommendation. *arXiv preprint*, 2026.

Peer-Reviewed Conference Publications

- [4] **Guanghui Min**, Tianhao Huang, Ke Wan, and Chen Chen. GIST: Targeted Data Selection for Instruction Tuning via Coupled Optimization Geometry. In *International Conference on Machine Learning (ICML)*, 2026.
- [5] **Guanghui Min**, Tianhao Huang, Ke Wan, Qi R. Wang, and Chen Chen. Bridging Dynamics and Data: A Unified Diffusion Framework for Mechanistically-Informed Epidemic Forecasting. In *International Conference on Machine Learning (ICML)*, 2026.
- [6] Tianhao Huang, **Guanghui Min**, Zhenyu Lei, Aiying Zhang, and Chen Chen. Uncovering Latent Communication Patterns in Brain Networks via Adaptive Flow Routing. In *International Conference on Machine Learning (ICML)*, 2026.
- [7] Haochen Liu, **Guanghui Min**, Song Wang, Yada Zhu, Chen Chen, and Jundong Li. LSP-RAG: Latent Scoring Perspectives for Knowledge Graph Retrieval-Augmented Generation. In *IEEE International Conference on Data Mining (ICDM)*, 2026.
- [8] **Guanghui Min**, Yinhan He, and Chen Chen. Scaling Epidemic Inference on Contact Networks: Theory and Algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- [9] Yinhan He, Chen Chen, Song Wang, **Guanghui Min**, and Jundong Li. Demystifying Epidemic Containment in Directed Networks: Theory and Algorithms. In *ACM International Conference on Web Search and Data Mining (WSDM)*, 2025.

Peer-Reviewed Workshop Publications

- [10] Ke Wan, Tianyi Zhao, Tianhao Huang, **Guanghui Min**, and Chen Chen. Calibrated Target-Aware Data Selection for Continual Mid-Training in Streams. In *Continual Adaptation at Scale: Towards Sustainable AI Workshop at International Conference on Machine Learning (ICML)*, 2026.
- [11] **Guanghui Min**, Wenxin Xu, Kateri DuBay, Chen Chen. Exploring Generative Approaches for Predicting Copolymer Sequences from Reaction Conditions. In *AI for Science Workshop at Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

WORK EXPERIENCE

2026/05 – 2026/08	Nokia of America <i>AI Research & Development Intern</i> Manager: Dr. Liang Wu	Sunnyvale, CA
	- Studied context compression for long-horizon LLM agents, including its effects on task success, step efficiency, and redundant information reacquisition. - Selected representative tool-use trajectories from raw agent rollouts for compressor training, failure diagnosis, and agent behavior analysis.	
2020/08 – 2024/07	Yinhua Fund Management Co., Ltd. <i>Senior Machine Learning Engineer, Innovation Center</i>	Beijing, China
	- Developed machine-learning workflows for investment research and portfolio analysis, including time-series clustering and graph-based methods for diversification and risk reduction. - Applied LLMs and generative models to investment research and wealth-management scenarios, including parameter-efficient adaptation with LoRA and P-tuning. - Built internal research-system modules for active-equity analysis, simulated portfolios, periodic reports, and workflow automation.	

HONORS & AWARDS

2025	NeurIPS Scholar Award
2024	UVA Computer Science Scholarship
2016–2018	Wuhan University Third-Class Scholarship (top 15%), Statistics
2015	Wuhan University Second-Class Scholarship (top 10%), Fundamental Mathematics

TEACHING EXPERIENCE

Teaching Assistant

Fall 2026	CS 6501-007: Graph Machine Learning <i>Instructor: Prof. Chen Chen</i>	University of Virginia
Spring 2026	CS 4774: Machine Learning <i>Instructor: Prof. Chen Chen</i>	University of Virginia
Fall 2025	CS 6205: Research Methods in Human-Computer Interaction <i>Instructor: Prof. Seongkook Heo</i>	University of Virginia

TECHNICAL SKILLS

Programming	Python, R, MySQL, Oracle, Redis
ML / LLM	PyTorch, Transformers, Hugging Face, LoRA, scikit-learn, TensorFlow
Research	LLM post-training, data-centric AI, long-horizon agents, time-series modeling, graph learning